



CORE CS (THEORY)

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❖ Top 50 Theory Questions for Interviews

1. What is Computer Science?

Computer Science deals with computation, data, and algorithms.
It focuses on problem-solving using computers.
It combines theory, design, and practical implementation.

2. What is an Algorithm?

An algorithm is a finite set of steps to solve a problem.
It must be correct, efficient, and terminate.
Algorithms are language-independent.

3. What is Data?

Data is raw facts and figures.
It becomes information after processing.
Data is the foundation of computing systems.

4. What is Information?

Information is processed and meaningful data.
It supports decision-making.
Accuracy and relevance are critical.

5. What is Software?

Software is a collection of programs and instructions.
It directs hardware to perform tasks.
Examples include system and application software.

6. System Software vs Application Software

System software manages hardware resources.
Application software serves user needs.
Examples are OS and word processors.

7. What is Hardware?

Hardware refers to physical components of a computer.
Includes CPU, memory, and peripherals.
It executes software instructions.

8. What is an Operating System?

An OS manages hardware and software resources.
It provides an interface between user and machine.
It ensures efficient system operation.

9. What is a Process?

A process is a program in execution.
It has its own memory and resources.
Processes are managed by the OS.

10. What is a Thread?

A thread is a lightweight execution unit.
Multiple threads can exist in one process.
They share memory space.

11. What is Multitasking?

Multitasking allows multiple tasks to run concurrently.
CPU time is shared among processes.
It improves system efficiency.

12. What is Memory?

Memory stores data and instructions.
It can be primary or secondary.
Speed and capacity vary by type.

13. What is Virtual Memory?

Virtual memory uses disk as extended RAM.
It allows large programs to run.
Managed by the OS.

14. What is a File System?

A file system organizes data on storage devices.
It controls file access and storage.
Examples include FAT and NTFS.

15. What is DBMS?

DBMS manages structured data.
It provides security and consistency.
Supports data retrieval and storage.

16. What is a Database?

A database is an organized data collection.
It enables efficient data access.
Used in applications and systems.

17. What is Normalization?

Normalization reduces data redundancy.
It improves data integrity.
Achieved using normal forms.

18. What is a Key in DBMS?

A key uniquely identifies records.
Ensures data consistency.
Examples include primary and foreign keys.

19. What is a Data Structure?

A data structure organizes data efficiently.
It optimizes access and modification.
Examples include arrays and trees.

20. What is an Array?

An array stores elements of same type.
Memory is contiguous.
Access is fast using index.

21. What is a Linked List?

A linked list stores nodes with pointers.
Dynamic in size.
Insertion is efficient.

22. Stack vs Queue

Stack follows LIFO.
Queue follows FIFO.
Used for different access patterns.

23. What is a Tree?

A tree is a hierarchical structure.
It has nodes and edges.
Used in searching and sorting.

24. What is a Graph?

A graph represents relationships.
Consists of vertices and edges.
Used in networks.

25. What is Recursion?

Recursion is function calling itself.
Requires base condition.
Uses stack memory.

26. What is Time Complexity?

Time complexity measures algorithm efficiency.
Depends on input size.
Expressed using Big-O.

27. What is Space Complexity?

Space complexity measures memory usage.
Includes auxiliary space.
Important for optimization.

28. What is Compiler?

A compiler translates code to machine language.
Execution is fast.
Errors found before execution.

29. Interpreter vs Compiler

Interpreter executes line by line.
Compiler translates entire program.
Execution differs in speed.

30. What is Programming Language?

A programming language is used to write programs.
It follows specific syntax and semantics.
Examples include C and Java.

31. What is OOP?

OOP organizes software using objects.
It improves modularity.
Supports reusability.

32. OOP Principles

Encapsulation, Inheritance, Polymorphism, Abstraction.
They improve design.
Enhance maintainability.

33. What is Encapsulation?

Encapsulation binds data and methods.
Restricts direct access.
Improves security.

34. What is Inheritance?

Inheritance enables reuse of existing code.
Child class derives parent features.
Supports hierarchy.

35. What is Polymorphism?

Polymorphism allows multiple behaviors.
Same interface, different implementation.
Improves flexibility.

36. What is Abstraction?

Abstraction hides complexity.
Shows essential features.
Improves usability.

37. What is Exception Handling?

Exception handling manages runtime errors.
Prevents abrupt termination.
Improves reliability.

38. What is Networking?

Networking connects systems.
Allows data sharing.
Uses protocols.

39. What is Protocol?

Protocol defines communication rules.
Ensures data consistency.
Examples include TCP/IP.

40. What is OSI Model?

OSI has seven layers.
Standardizes communication.
Used for troubleshooting.

41. What is Internet?

Internet is a global network.
Connects millions of devices.
Uses TCP/IP.

42. What is HTTP?

HTTP is a web protocol.
It is stateless.
Follows request-response model.

43. What is Cyber Security?

Cyber security protects systems.
Prevents unauthorized access.
Ensures data safety.

44. What is Encryption?

Encryption converts data into unreadable form.
Protects confidentiality.
Requires keys.

45. What is Software Engineering?

Software engineering applies systematic methods.
Ensures quality software.
Focuses on development lifecycle.

46. What is SDLC?

SDLC defines software development stages.
Includes planning to maintenance.
Ensures structured development.

47. What is Testing?

Testing verifies software correctness.
Finds defects.
Improves reliability.

48. What is Debugging?

Debugging removes errors.
Occurs after testing.
Improves performance.

49. What is Cloud Computing?

Cloud computing delivers services online.
Reduces infrastructure cost.
Offers scalability.

50. What is Artificial Intelligence?

AI enables machines to simulate intelligence.
Uses algorithms and data.
Applied in automation.

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